

Rating Municipal Bonds by AI-Crash Credit Sensitivity

Signal OS · muni_credit methodology · the β -CapGains framework · v1.4 (validated + blinded + count-hardened + CDIAC-checked) · 2026-06-06 · reproducible: `capgains_beta.py`

1

transmission channel
(state-GF cap-gains income tax)

-71%

calibration shock
(CA cap-gains rev \$17B→\$5B, dot-com)

5

rating tiers
AI-1 Insulated → AI-5 Direct

0-1+

β scale
(state GO $\approx$ 1.00)

1 · What this rates — and what it does not

A muni's **credit/spread sensitivity to a California capital-gains shock** — the channel through which an **AI-equity bubble burst** reaches muni credit (the same pipe the dot-com bust used). It is a **relative credit-stress rating**, not: (a) a *default* rating — CA GO debt service is constitutionally 2nd-priority, so default $\beta \approx 0$ even at AI-5; (b) a *total-return / price* rating — price also rides the rate regime (see §5 caveat). Hold rates constant; this isolates the cap-gains credit leg.

2 · The transmission model

California's General Fund is the most tech-levered state revenue base in the U.S. via **capital-gains + stock-option + IPO-wealth personal income tax** (avg ~7.9% of GF revenue, ~3× the volatility of overall PIT). An AI crash evaporates that wealth and the revenue with it. **Anchor:** in the dot-com bust this revenue ran **\$17B (2000-01) → \$5B (2002-03), -71%** [LAO]; **CA State GO was cut AA/Aa2→BBB/A2 (4-5 notches, worst state)** while SB-222 school GO and water revenue were **untouched**. The β scale is normalized so an **unshielded state-GF income-tax claim = 1.00**, calibrated to that ordering.

3 · The scoring formula

$\beta_{\text{capgains}} = \text{channel-elasticity} \times \text{structure-shield} \times \text{region-factor}$ (normalized: state-GF income-tax claim = 1.00)

Revenue-channel elasticity	value
state gf income tax	1.00
county gf appropriation	0.35
tax increment	0.20
muni revenue authority	0.15
ccrc entrance	0.30
hospital revenue	0.12
housing finance	0.10
school go av	0.05
water revenue	0.03

Credit-structure shield	value
state go	1.00
school go sb222	0.60
tab	0.90
hospital naked	1.00
ccrc calmortgage	0.45
water	0.95
calhfa	0.90
county cop	0.85
muni rev	0.95

Region factor: Silicon Valley / Bay Area = 1.40 · Bay-adjacent = 1.10 · elsewhere = 1.00. **Uncertainty:** ±40% band on the channel elasticity → per-name low/high.

4 · The rating scale

Tier	β range	Revenue archetypes	Dot-com behavior (validation)	n in basket
AI-1 Insulated	≤ 0.05	Statutory-lien property-tax GO (SB-222 school), water revenue	Untouched in dot-com	8
AI-2 Low	0.05 – 0.15	Enterprise revenue (hospital, CCRC-insured), housing finance	Minor spread drift	4
AI-3 Moderate	0.15 – 0.30	Tax-increment TAB, tech-hub hospital, county GF appropriation	1–2 notch / spread widening	7
AI-4 High	0.30 – 0.60	Local-GF appropriation, weak/late-base tax increment	Multi-notch in a deep shock	0
AI-5 Direct	> 0.60	State GF / income-tax-backed GO & appropriation	4–5 notches (CA AA→BBB, 2003)	1

5 • Modifiers & the critical caveat

- **Notch DOWN (more insulated):** statutory lien (SB-222), bond insurance / Cal-Mortgage-HCAI (state-credit substitution), dedicated non-GF pledge.
- **Notch UP (more exposed):** tech-hub region (direct tech-economy footprint), GF appropriation / lease structure, late-vintage tax-increment with a thin base above the frozen AV.

Credit ≠ price. This rating fixes rates. *Total-return* safety in an AI crash also depends on the **rate regime**: the muni hedge works only in a **Fed-cutting growth scare** (dot-com analog — long duration becomes a tailwind on top of insulated credit). In a **liquidity crunch** (2008, Mar-2020) or **inflation / rate-shock** (2022) it degrades, breaks, or inverts — munis fall *with* equities regardless of a clean AI-credit rating. A low β -CapGains is **necessary, not sufficient**, for drawdown safety.

6 • Worked example — CA Muni Honesty Basket (n=20)

Equal-weight basket $\beta = 0.16$ (band [0.095, 0.222]) — ~15% of the cap-gains credit-sensitivity of an all-state-GO portfolio → overall rating **AI-2 (Low)**, low end of Moderate when blended. **Concentration:** the single CA State GO drives **31.5%** of the entire basket β ; ex that name, $\beta = 0.115$. The inert tail (6 SB-222 school GO + 2 water) is 7.4% of total β across 8 of 20 names.

#	Obligor	Channel	β	Rating
19	California State General Obligation	state gf income tax	1.000	AI-5 Direct
20	County of San Diego COP (SUB for Pensi	county gf appropriation	0.297	AI-3 Moderate
14	Successor Agency to the San Jose Redev	tax increment	0.252	AI-3 Moderate
12	Anaheim Housing & Public Improvements	tax increment	0.180	AI-3 Moderate
15	Successor Agency to the Fontana Redeve	tax increment	0.180	AI-3 Moderate
16	Cloverdale CDA Successor Agency (TAB)	tax increment	0.180	AI-3 Moderate
7	El Camino Hospital (CHFFA conduit 1303	hospital revenue	0.168	AI-3 Moderate
8	Stanford Health Care (CHFFA conduit 13	hospital revenue	0.168	AI-3 Moderate
9	Aldersly (Garden Retirement Community)	ccrc entrance	0.149	AI-2 Low
13	Anaheim Housing & Public Improvements	muni revenue authority	0.142	AI-2 Low
10	Bethany Home Society of San Joaquin Co	ccrc entrance	0.135	AI-2 Low
11	CalHFA program bond	housing finance	0.090	AI-2 Low
1	Duarte Unified School District	school go av	0.030	AI-1 Insulated
2	Alpine County USD GO (SUB for Alpine U	school go av	0.030	AI-1 Insulated
3	Clovis Unified School District GO	school go av	0.030	AI-1 Insulated
4	Downey Unified School District GO	school go av	0.030	AI-1 Insulated
5	Sanger Unified School District GO	school go av	0.030	AI-1 Insulated
6	Pomona Unified School District GO	school go av	0.030	AI-1 Insulated
17	Metropolitan Water District of Souther	water revenue	0.028	AI-1 Insulated
18	Marina Coast Water District (Rev)	water revenue	0.028	AI-1 Insulated

7 · Backtest validation — does the rating predict reality?

Two out-of-sample tests (free-data feasible); a third proposed. **Both confirm the rating's central claims.**

7.1 Credit-tier dispersion is regime-conditional

High-yield muni (VWAHX — proxy for lower-credit / higher- β revenue & weaker structure) vs investment-grade long muni (VWLTX). Lower credit should lag — and most in *fundamental* busts, least in *rate* shocks.

Crash	Regime	HY muni (VWAHX)	IG long (VWLTX)	HY – IG
Dot-com 2000–02	growth scare, Fed cuts	+25.2%	+30.5%	–5.3 pp
GFC 2008	liquidity crisis	–10.5%	–4.9%	–5.6 pp
COVID 2020	liquidity shock (recovered)	+5.4%	+6.2%	–0.8 pp
2022	rate shock	–11.8%	–10.4%	–1.4 pp

Result: credit-tier dispersion is ~4× larger in the two fundamental busts (dot-com, GFC: ~–5.5 pp) than in the transient-liquidity / rate episodes (2020, 2022: ~–1 pp). β — credit-channel sensitivity — bites hardest in a real crisis, so a low- β rating is most protective exactly when it matters. *Caveat: HY-vs-IG is general credit quality, a loose proxy for the cap-gains channel specifically.*

7.2 CA State GO credit tracks cap-gains, not rates (the AI-5 anchor)

The tight, on-point test: if $\beta=1.0$ for state GO is right, CA GO credit should move with the cap-gains cycle and be *orthogonal* to rate shocks.

Episode	Cap-gains revenue	CA GO rating action	Validates
Dot-com 2001–03	\$17B → \$5B (–71%)	AA → BBB (–4–5 notches)	AI-5 ✓
GFC 2008–09	→ ~\$2B (2009 trough)	A → Baa1/BBB + IOUs	AI-5 ✓
Recovery 2010s–’21	rebuilt (record 2021)	upgraded → Aa2/AA	symmetric ✓
2022 rate shock	still high	no downgrade	credit≠price ✓

Result: CA State GO was slashed in **both equity-bubble busts** (cap-gains collapse), recovered **symmetrically** as cap-gains rebuilt, and was **untouched in the 2022 rate shock** — confirming the cap-gains channel drives GO credit (validates AI-5 = 1.0) and that the 2022 muni drawdown was pure price/duration (validates the credit≠price caveat).

7.3 Cross-sectional ordering test — blinded agents

Does the full AI-1...AI-5 scale rank-order by *actual* credit stress in each crash? **Blinded method:** the β -tier sector order was sealed to disk *first*; four research agents then measured each sector’s realized credit outcome (ratings, defaults, spreads) per crash with **no knowledge** of the β -framework, the cap-gains hypothesis, the AI-crash framing, or the expected order. Spearman rank-corr of their blind rankings vs the sealed prediction:

Crash (shock type)	Spearman ρ	State GO actual rank	Read
Dot-com 2000–03 (income / cap-gains)	0.90	#1 (most stressed)	strong — the AI archetype
GFC 2008–09 (property / liquidity)	0.33	#3	diverges: CCRC/CalHFA jump up
COVID 2020 (operational)	0.45	#5	weak: operational sectors top
2022 (rate / inflation)	0.24	#6 (upgraded)	State GO upgraded — cap-gains was high

Result — the rating is channel-SPECIFIC, exactly as designed. It rank-orders credit stress at **$\rho=0.90$ in the income/cap-gains regime** (the dot-com bust = the AI-crash archetype) and *correctly* does NOT predict the property (2008), operational (2020), or rate (2022) regimes — a generic credit-quality proxy would score high everywhere. The **State GO is the tell:** most-stressed (#1) when cap-gains collapsed (dot-com), but **upgraded** (#6) in 2022 when cap-gains hit a record — the AI-5 name swings with the cap-gains cycle, confirming the mechanism. The **insulated tier (school GO, water) sat bottom-ranked in all four regimes.** *Caveat: dot-com anchored the scale’s endpoints, so its ρ is out-of-sample in the middle tiers but not the extremes; the cleaner OOS evidence is the discriminant pattern across the other three crises. Agents’ middle-tier (county COP, TAB) rankings were often low-confidence on thin data.*

7.4 Hardening with per-issuer counts (blinded round 2)

A second blinded round tried to replace the agents' middle-tier *judgment* with *enumerated* rating-action counts (rubric sealed first). **Decisive finding — a data limit:** per-issuer CA rating counts for the middle tiers are **not publicly recoverable** (proprietary Moody's/S&P/Fitch DBs + CDIAC default-draw index); defaults are enumerable, mid-tier downgrades are not. Scoring what *is* documented: **hardened dot-com $\rho = 0.80$** (vs 0.90 judgment-based — five middle sectors tie at zero documented CA actions, so the middle is *unresolved, not wrong*); the **insulated tier had 0 documented actions in all four regimes**; State GO took ~9 multi-notch actions; CalHFA was upgraded in 2022. **The rating is robust at the extremes and bucket level; the fine 8-way middle ordering is data-limited, not validated.** *Refinement (model-v2):* TAB/hospital/CCRC showed their stress in the property/operational regimes, not income — so β -CapGains is conservative for the cap-gains channel; re-classing them toward AI-1/2 for income shocks would raise the fit.

CDIAC index — scripted, proven dead-end for this. The one CA-specific granular source (CDIAC default-draw) was pulled in full via the real DebtWatch API (710 events, 1991–2026). It **cannot harden the eight sectors**: it is **89% Mello-Roos/CFD land-secured draws**, and the rating-test sectors appear 0–3× each — because GO/school/hospital/CCRC/water/county stress shows up as *downgrades*, which CDIAC does not track (rating actions live only in proprietary agency archives). The public-data limit is now demonstrated, not asserted.

(Reproducible: `cdiac_draws.py`.)

8 • Caveats

Structural model, not a regression — per-obligor 20-yr pledged-revenue series don't exist to regress on CA cap-gains receipts; elasticities are first-principles, anchored to the dot-com episode + LAO data. **CA-specific** — generalizes to other states by their GF income-tax / cap-gains share (CA is the high extreme; TX/FL with no income tax → near-zero state channel). **Single-name concentration** dominates a small basket — read β at the name level, not just the average. **β is a spread/rating beta**, calibrated to relative ordering, not a forecast of basis points.

Sources: LAO (CA capital-gains volatility & GF share); CA State Treasurer GO ratings history; S&P / Moody's / Fitch dot-com rating actions; Signal OS muni_credit holdings & `capgains_beta.py`. Methodology rates relative AI-crash credit sensitivity for analyst use; not investment advice.